



## DEVELOPMENT

# Pump it Up: Data Mining the Water Table

Can you predict which water pumps are faulty to promote access to clean, potable water across Tanzania? This is an intermediate-level practice competition.

**Intermediate** practice**1 year** left**19,827** joined[Compete! →](#)

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## Problem description

This is where you'll find all of the documentation about this dataset and the problem we are trying to solve.

## The features in this dataset

Your goal is to predict the operating condition of a waterpoint for each record in the dataset. You are provided the following set of information about the waterpoints:

- `amount_tsh` - Total static head (amount water available to waterpoint)
- `date_recorded` - The date the row was entered
- `funder` - Who funded the well
- `gps_height` - Altitude of the well
- `installer` - Organization that installed the well
- `longitude` - GPS coordinate
- `latitude` - GPS coordinate
- `wpt_name` - Name of the waterpoint if there is one
- `num_private` -
- `basin` - Geographic water basin
- `subvillage` - Geographic location
- `region` - Geographic location
- `region_code` - Geographic location (coded)
- `district_code` - Geographic location (coded)
- `lga` - Geographic location
- `ward` - Geographic location
- `population` - Population around the well
- `public_meeting` - True/False
- `recorded_by` - Group entering this row of data
- `scheme_management` - Who operates the waterpoint
- `scheme_name` - Who operates the waterpoint
- `permit` - If the waterpoint is permitted
- `construction_year` - Year the waterpoint was constructed
- `extraction_type` - The kind of extraction the waterpoint uses
- `extraction_type_group` - The kind of extraction the waterpoint uses
- `extraction_type_class` - The kind of extraction the waterpoint uses
- `management` - How the waterpoint is managed
- `management_group` - How the waterpoint is managed
- `payment` - What the water costs
- `payment_type` - What the water costs
- `water_quality` - The quality of the water
- `quality_group` - The quality of the water
- `quantity` - The quantity of water
- `quantity_group` - The quantity of water
- `source` - The source of the water
- `source_type` - The source of the water

- **source\_class** - The source of the water
- **waterpoint\_type** - The kind of waterpoint
- **waterpoint\_type\_group** - The kind of waterpoint

## Feature data example

For example, a single row in the dataset might have these values:

|                      |                    |
|----------------------|--------------------|
| <b>amount_tsh</b>    | 300.0              |
| <b>date_recorded</b> | 2013-02-26         |
| <b>funder</b>        | Germany Republi    |
| <b>gps_height</b>    | 1335               |
| <b>installer</b>     | CES                |
| <b>longitude</b>     | 37.2029845         |
| <b>latitude</b>      | -3.22870286        |
| <b>wpt_name</b>      | Kwaa Hassan Ismail |
| <b>num_private</b>   | 0                  |
| <b>basin</b>         | Pangani            |
| <b>subvillage</b>    | Bwani              |
| <b>region</b>        | Kilimanjaro        |
| <b>region_code</b>   | 3                  |

|                              |                             |
|------------------------------|-----------------------------|
| <b>district_code</b>         | 5                           |
| <b>lga</b>                   | Hai                         |
| <b>ward</b>                  | Machame Uroki               |
| <b>population</b>            | 25                          |
| <b>public_meeting</b>        | True                        |
| <b>recorded_by</b>           | GeoData Consultants Ltd     |
| <b>scheme_management</b>     | Water Board                 |
| <b>scheme_name</b>           | Uroki-Bomang'ombe water sup |
| <b>permit</b>                | True                        |
| <b>construction_year</b>     | 1995                        |
| <b>extraction_type</b>       | gravity                     |
| <b>extraction_type_group</b> | gravity                     |
| <b>extraction_type_class</b> | gravity                     |
| <b>management</b>            | water board                 |
| <b>management_group</b>      | user-group                  |
| <b>payment</b>               | other                       |
| <b>payment_type</b>          | other                       |
| <b>water_quality</b>         | soft                        |

|                       |                    |
|-----------------------|--------------------|
| quality_group         | good               |
| quantity              | enough             |
| quantity_group        | enough             |
| source                | spring             |
| source_type           | spring             |
| source_class          | groundwater        |
| waterpoint_type       | communal standpipe |
| waterpoint_type_group | communal standpipe |

## The labels in this dataset

The labels in this dataset are simple. There are three possible values:

- **functional** - the waterpoint is operational and there are no repairs needed
- **functional needs repair** - the waterpoint is operational, but needs repairs
- **non functional** - the waterpoint is not operational

## Submission format

The format for the submission file is simply the row id and the predicted label (for an example, see [SubmissionFormat.csv](#) on the data download page).

For example, if you just predicted that all the waterpoints were functional you would have the following predictions:

| id    | status_group |
|-------|--------------|
| 50785 | functional   |
| 51630 | functional   |
| 17168 | functional   |
| 45559 | functional   |
| 49871 | functional   |

Your `.csv` file that you submit would look like:

```
id,status_group
50785,functional
51630,functional
17168,functional
45559,functional
...
```

## Good luck!

Good luck and enjoy this problem! If you have any questions you can always visit the [user forum](#)!

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